

Final Report: Cybersecurity Log Analysis with HDFS, Spark, and Streamlit

Project Goals:

The primary goal of this project was to develop a local data pipeline that leverages Hadoop Distributed File System (HDFS), YARN, and Apache Spark to analyze a cybersecurity dataset. The output of this analysis is then presented through a web-based interface built with Streamlit. This project demonstrates the ability to process and visualize large-scale data using distributed computing tools on a single-node setup.

Functionalities:

- Store cybersecurity logs (cyberlogs.csv) in HDFS
- Process the log data using PySpark and Apache Spark
- Use YARN as the resource manager for Spark jobs
- Perform simple data aggregations and column selections in PySpark before presenting them in the frontend
- Present data through a Streamlit web application
- Allow end-users to view and explore the logs via a web interface

Architecture and design:

This project follows a modular, single-node architecture built entirely on a Linux machine. The key components are:

HDFS (Hadoop Distributed File System): Used for storing the CSV data file reliably and in a scalable way.

YARN (Yet Another Resource Negotiator): Handles resource management and job scheduling for Spark tasks.

Apache Spark: Performs data loading, transformation, and analysis on the CSV dataset.

PySpark and Streamlit App: The project integrated PySpark data analysis and interactive Streamlit visualizations within a single Python file. This unified application read and analyzed cybersecurity logs from HDFS using Spark and then rendered dynamic charts and tables via Streamlit.

Data Flow

1. Data is uploaded to HDFS using Hadoop commands.
2. Spark reads the data from HDFS mainly using a PySpark script.
3. The PySpark Script performs a limited analysis (e.g., schema inspection, record filtering, Data Aggregation).
4. The results are passed to the Streamlit application and charts get displayed.

Discussion of the results:

This project successfully demonstrated the full pipeline of reading data from HDFS, Using YARN as a resource manager for Spark, analyzing the data with PySpark, and presenting the output in a Streamlit web application—managed from a single Python file. This approach allowed for good use of Big Data technologies and for very quick and rapid prototyping.

Key results include:

The Streamlit app accurately reflected the structure and insights from our cybersecurity log dataset, including column-wise breakdowns, data visualizations, and security insights.

Our PySpark script's ability to read and process the cyberlogs.csv file directly from HDFS proved efficient and uncomplicated.

The cybersecurity log dataset revealed several insights through PySpark analysis. These findings were visualized in Streamlit to show quick data visuals.

Performance was stable for a dataset of around 40,000 rows, with minimal latency in our pipeline and rendering outputs on the frontend.

Documentation of your project:

Deployment instructions and instructions to run the application are shown the the Github ReadMe

Github: <https://github.com/Zarifzz/hdfs-spark-project>